

# Continuous Grip-Force Dynamics as an Online Biophysical Signal of Perceptual Strain

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**Abstract** Older adults frequently perform perceptual decisions while sustaining physical effort, yet most dual-task research treats motor output as a binary manipulation rather than a continuous behavioral signal. Here we ask whether sub-second grip-force dynamics predict trial-level perceptual accuracy in older adults performing auditory and visual same/different discrimination while maintaining low (5%) or high (40%) isometric handgrip force scaled to maximum voluntary contraction. We extracted event-locked spectral power from the continuous dynamometer stream in a visuomotor tracking band (0.5–3.0 Hz) and a physiological tremor band (8.0–12.0 Hz), then tested whether these features explain perceptual accuracy beyond nominal task difficulty using mixed-effects logistic regression and subject-wise cross-validation. Physical load compressed perceptual sensitivity, and tremor-band power showed a numerically negative association with accuracy under high grip (40% MVC), although trial-level effects did not reach conventional significance after controlling for difficulty. Cross-validated models confirmed a modest incremental predictive gain from force features beyond task labels ( $\Delta\text{AUC} \approx 0.005$ ). Exploratory LC microstructure analyses showed non-significant directional trends only.

**Keywords:** aging, dual-task, handgrip force, perceptual decision-making, physiological tremor, cognitive-motor interference

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## Introduction

Daily life rarely separates cognition from action. Older adults often have to sustain force while selecting, monitoring, or discriminating information: carrying groceries while checking traffic, gripping a railing while navigating a crowded space, opening medication packaging while reading dosage instructions, or holding a suitcase while searching for a gate sign. These situations are not simply motor tasks performed beside cognitive tasks. They are concurrent control problems in which the nervous system must regulate force, sensory evidence, attention, and response selection at the same time. This problem becomes more consequential in later life because aging is associated with declines in processing speed, executive control, sensory precision, and neuromodulatory regulation, all of which can reduce the capacity to manage simultaneous physical and cognitive demands [1–5].

The dual-task literature has long shown that cognitive and motor demands interact in aging. Older adults show greater performance costs when walking while remembering, maintaining posture while performing a cognitive task, or coordinating fine motor output while holding information in working memory [6–12]. These effects are often interpreted as evidence that motor control and cognition draw on partially shared executive resources. The implication is straightforward but important: motor behavior in older adulthood cannot be treated as purely peripheral, and cognitive performance cannot be fully

understood when the physical state of the body is ignored. Even a simple sustained contraction may alter the resources, arousal state, and control processes available for perceptual decision-making.

Most cognitive-motor aging studies have focused on gait, balance, or posture. These paradigms are highly relevant for fall risk and functional mobility, but they do not exhaust the range of everyday physical-cognitive demands. Many daily actions are stationary rather than locomotor. A person may need to hold a bag, grip a steering wheel, stabilize a handrail, squeeze a bottle cap, or maintain force on an object while simultaneously interpreting sensory information. Isometric handgrip provides a controlled way to study this class of behavior. It allows physical effort to be scaled to each participant's own maximum voluntary contraction, minimizes whole-body movement, and can be used in constrained environments such as MRI scanning. It also has ecological relevance because power grip is involved in many everyday actions requiring sustained force. Unlike gait or postural control, handgrip permits direct measurement of the force signal with high temporal resolution, allowing the motor task itself to become a continuous behavioral readout rather than a binary manipulation.

A central distinction for the present study is the difference between trait-level motor capacity and online motor regulation. Maximum voluntary isometric contraction, or MVIC, captures a relatively stable between-subject mea-

sure of grip strength. Grip strength has been widely used as an index of neuromuscular health and physical frailty in older adults [13–16]. Although grip strength partly reflects peripheral muscle properties, it is also shaped by neural factors involved in force generation, coordination, and motor control [17–20]. Recent work from our group extended this framework by using connectome-based predictive modeling to show that task-based functional connectomes can predict individual differences in MVIC in older adults, with subcortical and cerebellar networks playing an important role in the prediction of grip strength [21]. That work established MVIC as a meaningful trait-like marker linked to distributed brain organization.

However, a stable force ceiling does not answer a different question: how does the aging brain regulate force from moment to moment while a concurrent cognitive decision is unfolding? MVIC tells us how hard a person can squeeze. It does not tell us whether the person can hold a target force smoothly, whether the force signal destabilizes when perceptual evidence becomes difficult, or whether subtle oscillations in force output predict cognitive failure on the same trial. These are process-level questions. They require analysis of continuous motor dynamics rather than static motor capacity. This distinction is especially important in aging because an older adult may show preserved maximum strength but still struggle with the online regulation of force under cognitive load. Conversely, a participant with lower MVIC may maintain stable force if the motor-control loop remains well regulated. The present study therefore shifts the focus from grip strength as a between-subject trait to grip-force dynamics as an online signal of cognitive-motor strain.

Two theoretical frameworks motivate this shift. The first is Resource Competition theory. In its classic form, capacity models propose that attention and cognitive control depend on a limited pool of processing resources that must be allocated across concurrent demands [22–24]. When two tasks are performed at once, performance costs arise because both tasks require access to overlapping resources. Maintaining handgrip force is not just a muscular event. It requires sustained goal maintenance, visual feedback monitoring, error correction, and descending motor control. Perceptual discrimination similarly requires sensory encoding, evidence accumulation, response selection, and often the suppression of irrelevant information. Under low physical and cognitive demand, these operations may proceed in parallel with little cost. Under high physical load and high perceptual

difficulty, however, the combined demand may exceed the available control capacity, especially in older adults who already operate closer to processing limits [1, 3, 4].

Resource Competition theory also highlights why aging should amplify cognitive-motor interference. Older adults often show reduced efficiency in allocating attention across competing inputs and weaker suppression of irrelevant information [2, 5]. Inhibitory-control deficits are especially relevant because concurrent physical effort may increase the need for control at the same time that the perceptual task requires discrimination of subtle sensory differences. A central-resource account would therefore predict that increasing physical effort should reduce the resources available for perceptual evidence accumulation and decision formation. It would also predict that force regulation itself may suffer when attention is withdrawn from the motor loop. From this perspective, instability in grip force is not a nuisance artifact. It is a measurable expression of resource reallocation under concurrent demand.

The second framework is the locus coeruleus–norepinephrine Adaptive Gain account. The LC-NE system regulates arousal, attention, and neural gain by modulating the responsiveness of cortical and subcortical processing systems [25–29]. Gain modulation changes the signal-to-noise properties of neural processing by amplifying strongly activated representations and suppressing weaker activity patterns [30]. In adaptive conditions, phasic LC-NE responses can sharpen processing and support task engagement. But the same system can become maladaptive when arousal is too high, too tonic, or poorly regulated. This logic follows the long-standing inverted-U relation between arousal and performance, in which moderate arousal can improve performance but excessive arousal degrades it [25, 31].

Effortful muscular contraction is a plausible route into this arousal system. Isometric handgrip can increase sympathetic activation, cardiovascular response, and noradrenergic engagement, making it useful for manipulating arousal without requiring locomotion [32–34]. Under moderate load, this arousal increase may improve processing speed or readiness. Under high load, especially when paired with difficult perceptual decisions, arousal may push the system beyond its optimal operating range. In aging, this possibility is particularly important because the LC-NE system is vulnerable to structural and functional changes across later life, and age-related alterations in arousal regulation may affect both perception and cognitive control [27, 35, 36]. Thus, the same physical

effort that increases alertness in one context may destabilize performance in another.

Recent handgrip studies support this dual-mechanism view. Park H-B, Ahn S, Zhang W [37] found that visual search under high physical effort was faster but more vulnerable to distractor interference. This pattern is theoretically informative because it suggests that effort-induced arousal can speed some aspects of processing while simultaneously weakening top-down inhibitory control. Azer L, Xie W, Park H-B, Zhang W [38] extended this logic to aging by showing that effortful handgrip impaired older adults' working-memory performance when distractors were present, consistent with the view that older adults are especially vulnerable when physical effort and cognitive control demands converge. Together, these studies argue against a simple account in which physical effort is uniformly beneficial or uniformly harmful. Instead, physical effort appears to alter both arousal and control. Whether the observed effect is positive, negative, or mixed depends on load, task demands, age, and the specific behavioral measure being analyzed.

This distinction matters for the present study because perceptual decision-making is not a single process. Auditory and visual discrimination require the nervous system to compare a standard stimulus with a probe stimulus, detect whether a change occurred, and commit to a same/different response. As stimulus similarity increases, evidence becomes weaker and the decision becomes more demanding. In this context, high physical effort may interfere with perception through at least two routes. A Resource Competition mechanism would predict that force maintenance competes with perceptual evidence accumulation for central resources. An Adaptive Gain mechanism would predict that effort-induced arousal changes the neural signal-to-noise ratio, potentially improving processing at moderate levels but impairing discrimination when combined demand becomes too high. These mechanisms are not mutually exclusive. They may operate together, and their combined expression may be visible in the motor signal itself.

Isometric grip control is especially well suited for detecting this interaction because maintaining a target force is an active closed-loop process. Participants must continuously compare current force with the target, adjust motor output, and suppress deviations from the desired force level. These adjustments depend on sensory feedback, motor prediction, corticospinal output, subcortical regulation, and cerebellar error correction. Neural systems involved in force production and relaxation include mo-

tor, premotor, basal ganglia, and cerebellar circuits [18, 39]. Grip-force tracking is also sensitive to feedback precision and task constraints [40]. As fatigue or effort increases, force output can become more variable and less stable, reflecting changes in both peripheral neuromuscular state and central motor drive [41–43].

For this reason, whole-trial averages are too blunt for the current question. Mean force can confirm whether participants generally squeezed near the target, but it cannot reveal how force control fluctuated around the moment of perceptual challenge. Similarly, average absolute error collapses across distinct sources of motor variability. A participant may show slow corrective drift, high-frequency tremor, or both; these patterns may have different physiological meanings. Frequency-domain analysis allows us to separate these components. By treating the 100 Hz dynamometer stream as a continuous time series, we can examine the spectral structure of force deviations within each trial and ask whether specific oscillatory features predict perceptual performance.

The present study focuses on two engineered frequency bands with distinct biophysical interpretations. The first is a low-frequency visuomotor tracking band from 0.5 to 3.0 Hz. Power in this range captures slower corrective adjustments around the target force. These fluctuations likely reflect the participant's ongoing attempt to maintain the target centerline using visual feedback and voluntary motor correction. A reduction in this band after stimulus presentation may indicate a withdrawal of attention from the motor-control loop, as resources are redirected toward perceptual evidence accumulation. In other words, a collapse in tracking-band power may signal that the participant is no longer actively correcting force output with the same degree of moment-to-moment control.

The second band is a high-frequency physiological tremor band from 8.0 to 12.0 Hz. Power in this range is expected to capture faster oscillatory components of the force signal associated with physiological tremor, neuromuscular fatigue, and arousal-linked motor instability. This band is theoretically important because it may reflect the somatic cost of combined physical and cognitive demand. If high physical effort increases sympathetic arousal and destabilizes motor output, then tremor-band power should increase under conditions of greater strain. More importantly, if this tremor increase reflects a breakdown in the system's ability to regulate concurrent demands, then it should predict perceptual choice failure

beyond nominal task labels such as grip level or stimulus difficulty.

A key methodological issue is that low-effort and high-effort grip trials should not be assumed to reflect the same motor regime. At 5% MVC, participants are close to the lower bound of voluntary force production, where small precision adjustments and target-normalization effects can dominate the measured signal. At 40% MVC, participants are under meaningful physical strain, and the same spectral features are more likely to reflect effort-linked motor instability. Pooling low- and high-grip trials can therefore mix qualitatively different sources of variance. This is not only a statistical concern. It is a theoretical concern. If low-effort precision noise and high-effort somatic instability are collapsed into a single model, the resulting coefficient may obscure the very cognitive-motor interaction the study is designed to detect. For that reason, the present study treats grip level as defining distinct motor-control regimes and places primary inferential emphasis on the high-effort condition, where tremor-band dynamics are most directly interpretable as a marker of physical-cognitive strain.

This stratified approach also addresses a broader limitation of nominal task designs. Traditional dual-task analyses often compare low versus high physical effort and low versus high cognitive effort, then ask whether the interaction predicts mean accuracy or reaction time. That approach is useful, but it treats effort as an external label rather than an observed physiological process. Two participants assigned to the same 40% MVC condition may show different levels of force stability, tremor, fatigue, and corrective control. Even within the same participant, some trials may be regulated smoothly while others show transient destabilization. Continuous grip-force sensing makes it possible to move beyond the nominal condition label and measure the state of the motor system on each trial.

The present study asks whether sub-second fluctuations in concurrent grip-force control can be resolved at the single-trial level during perceptual decision-making, and whether those fluctuations carry incremental information about accuracy beyond nominal task difficulty labels. We examine older adults performing auditory and visual standard-probe perceptual discrimination tasks while maintaining either low or high isometric handgrip force. Rather than reducing grip performance to a trial average, we extract event-locked force dynamics from the continuous dynamometer stream and quantify spectral power in the visuomotor tracking and physiological tremor

bands. We then test whether these motor features explain perceptual accuracy after accounting for task difficulty and subject-level structure—under conditions where difficulty and between-subject baselines are expected to dominate variance.

We advance two primary hypotheses. The **Visuomotor Collapse Hypothesis** predicts that power in the 0.5–3.0 Hz tracking band will drop following probe presentation on difficult trials, suggesting that perceptual challenge temporarily withdraws attentional resources from the motor-control loop and reduces active correction around the target force. The **Tremor Surging Hypothesis** predicts that, under high physical strain, power in the 8.0–12.0 Hz tremor band will increase as perceptual difficulty rises and will predict subsequent choice failures, supporting the interpretation that high-frequency force instability is an online biophysical signature of cognitive-motor strain rather than peripheral motor noise.

To test these hypotheses, we analyze a cohort of older adults who completed auditory and visual perceptual discrimination challenges while producing handgrip force scaled to their own MVIC. The task design parametrically varied perceptual difficulty, allowing accuracy to be modeled across graded stimulus conditions rather than only coarse difficulty bins. The analytic pipeline uses target-normalized force error, event-locked windowing, frequency-domain feature extraction, and mixed-effects models with subject-level structure. Because the central question is whether subtle motor dynamics can be parsed at all under strong task constraints, we pair in-sample mixed-effects models with subject-wise cross-validated prediction. This structure asks whether force features generalize beyond participant-level baselines—not whether they overwhelm the psychometric effects of stimulus difficulty.

The contribution of this study is therefore threefold. First, it extends prior work on grip strength and frailty by moving from trait-level MVIC to online force-control dynamics. Second, it links Resource Competition and Adaptive Gain accounts by treating continuous motor instability as a potential behavioral trace of both resource reallocation and arousal dysregulation. Third, it introduces a stratified signal-processing framework for testing when high physical effort may transform grip-force tremor into a trial-level marker of perceptual strain—a question that requires separating load-induced instability from low-effort precision noise. In doing so, the study reframes the dynamometer from a device used only to impose physical effort into a high-frequency behavioral



sensor for probing subtle cognitive-motor coupling, even when that coupling is small relative to task difficulty.

## Methods

### Participants and procedure

Seventy older adults were recruited from communities surrounding the University of California, Riverside through mail, flyers, social media, and word-of-mouth. Inclusion criteria required participants to be between 60 and 90 years of age, have normal hearing and normal or corrected-to-normal vision, and be able to understand English well enough to follow task instructions. Participants were required to be right-handed and free of MRI contraindications, psychotropic medications, significant neurological conditions, and major health conditions such as diabetes. An a priori power analysis using G\*Power 3.1 [44] indicated that a sample of 34 participants would provide 80% power ( $\alpha = .05$ ) to detect a medium within-subject effect ( $f = 0.25$ ) in a repeated-measures design with two grip levels and an assumed correlation of 0.5. Two participants were removed prior to analysis — one for a possible diagnosis of mild cognitive impairment and one for a possible diagnosis of essential tremor — leaving a behavioral sample of 66 participants. Additional trial-level data cleaning (described below) left a final sample of 65 participants who completed both tasks (mean age = 69.5 years,  $SD = 6.4$  years; 50% female). All participants provided written informed consent and were compensated. All procedures were approved by the University of California, Riverside Institutional Review Board. The study was not pre-registered.

The present grip-force spectral analysis is restricted to participants for whom continuous dynamometer recordings were of sufficient quality and duration for frequency-domain decomposition. After applying grip-adherence and signal-quality screens described below, the analytic sample comprised 65 participants contributing 13,759 analysis-ready trials across the auditory and visual tasks. Full demographic characteristics are reported in Sun AY, Dastgheib M, Hu X, et al [45].

Participants completed four sessions. Sessions 1, 2, and 3 were conducted on separate days in the laboratory; Session 4 was administered remotely via tablet and contained neuropsychological assessments not reported here. **Session 1** introduced the tasks and established each participant's maximum voluntary contraction (MVC). Participants first watched instructional videos for the auditory and visual tasks without grip and completed at

least 60 practice trials per task. MVC was measured using three 5-second maximum squeezes with approximately 30 seconds of rest between trials; the mean force across the three trials was taken as MVC. Individual target forces were set at 5% MVC (low grip) and 40% MVC (high grip). Participants then practiced both tasks with grip cues and confidence ratings included, and received corrective feedback if grip targets were not met. **Sessions 2 and 3** each consisted of one task performed inside the MRI scanner (functional MRI data are not reported here), with task order counterbalanced across participants. MVC was re-measured at the start of Session 2 inside the scanner, and this value was used to normalize target grip force for both scanning sessions. Participants completed a short practice block outside the scanner before each scanning session, followed by five blocks of 30 trials per task.

### Task design and apparatus

Participants performed separate auditory and visual same/different perceptual discrimination tasks. On each trial, a standard stimulus and a probe stimulus were presented in succession, and participants judged whether the two stimuli were the same or different; they then provided a retrospective confidence rating on a 4-point scale (1 = low, 4 = high). The task structure is illustrated in Figure 1.

**Trial structure.** Each trial began with a 1.5–4.5 s jittered blank screen, followed by a 3 s display of a grip gauge instructing either low (5% MVC) or high (40% MVC) effort, with a color-coded fill providing real-time feedback (green = acceptable range, yellow = slight deviation, red = major deviation). A 0.25 s blank and a 0.5 s fixation dot preceded the stimuli. The standard stimulus was presented first, followed by a 0.5 s interstimulus interval (ISI), then the probe. Total trial duration was approximately 10.7 s. After probe offset, participants released the grip and had 3 s to report same or different (right index finger = “different,” right middle finger = “same”), followed by a 3 s confidence rating window. Grip level was varied pseudorandomly across trials in equal proportions.

**Auditory task.** The standard stimulus was a 0.1 s sinusoidal tone at 1000 Hz. The probe was either identical to the standard (0 Hz offset) or positively offset by 8, 16, 32, or 64 Hz.

**Visual task.** The standard was a centrally presented sinusoidal grating (Gabor patch) at 1.5 cycles per degree, Michelson contrast 0.2, and 4 degrees of visual angle. The

probe was either identical or positively offset in contrast by 0.06, 0.12, 0.24, or 0.48.

Each task comprised 150 trials split across five blocks, with each of the five probe levels presented in equal proportions. Stimulus generation and presentation were controlled using Psychophysics Toolbox Version 3.0.18 [46, 47] in MATLAB R2022b.

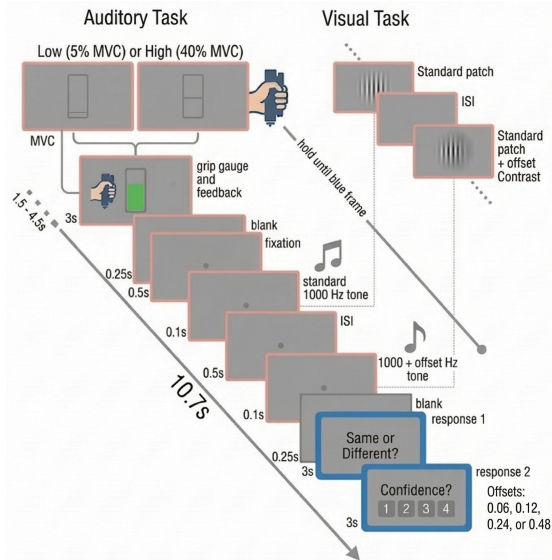


Figure 1: Trial structure for the auditory (left) and visual (right) tasks. Each trial began with a jittered blank period (1.5–4.5 s), followed by a 3 s grip gauge display instructing low (5% MVC) or high (40% MVC) force with real-time color feedback. A 0.25 s blank and 0.5 s fixation preceded the standard–probe stimulus pair (0.5 s ISI). After probe offset participants released the grip, then made a same/different judgment (3 s) and a 4-point confidence rating (3 s). Total trial duration ~10.7 s.

The experimental task was programmed and run on an Alienware laptop. Visual stimuli were projected using a PROPixx projector (VPixx Technologies, Montreal, Canada; 480 Hz refresh rate, 1920 × 1080 resolution). Participants viewed stimuli through a mirror mounted on the MRI head coil at an effective viewing distance of 12 cm. Auditory stimuli were delivered via MRI-safe earbuds (30 dB NRR attenuation; Newmatic Medical). Grip force was recorded continuously at 100 Hz using a Current Designs Grip Force dynamometer (Current Designs Inc., Philadelphia, USA). Participant responses were collected via a RESPONSEpixx fiber-optic response box (VPixx Technologies). Pupil diameter was recorded binocularly using a TRACKPixx MRI-compatible eye tracker (VPixx Technologies, 2000 Hz; pupillometry data are not reported here).

## Data processing and quality control

The present study treats the 100 Hz dynamometer stream as a continuous behavioral time series rather than a per-trial aggregate. All signal processing was implemented in Python 3 using SciPy [48].

**Epoch extraction.** For each trial, the continuous force record was segmented into a window beginning at grip-cue onset and extending 1 s after probe stimulus onset. This window captures both the preparatory holding period and the motor state at the moment of perceptual challenge. Reaction time was locked to probe onset, consistent with prior analyses of the same dataset [49].

**Target normalization.** Raw force values were divided by each participant’s MVC-derived target force for that trial (5% MVC or 40% MVC), expressing deviations as a proportion of the instructed level. This normalization ensures that spectral features reflect tracking quality relative to the intended effort level rather than between-subject differences in absolute grip strength.

**Detrending and filtering.** Each epoch was linearly detrended to remove slow drift. A 4th-order zero-phase Butterworth high-pass filter with a 0.5 Hz cutoff was then applied using `sosfiltfilt` to attenuate macro-level drift below the analysis bands while preserving the oscillatory structure of interest.

**Spectral estimation.** Power spectral density was estimated for each trial epoch using Welch’s method with a Hann window, 50% overlap, and a frequency resolution of approximately 0.5 Hz. PSD was log-transformed prior to modeling to reduce skew.

**Band-power features.** Two frequency bands were extracted from each trial’s PSD:

- *Visuomotor tracking band* (0.5–3.0 Hz): average log-power reflecting slow corrective adjustments to keep force near the target. A reduction in this band after probe onset may indicate withdrawal of attentional resources from the motor-control loop.
- *Physiological tremor band* (8.0–12.0 Hz): average log-power reflecting higher-frequency neuromuscular instability, sensitive to fatigue and arousal-linked perturbations of force output. An increase in this band under high load is interpreted as the somatic expression of overdriven motor control.

**Stratification by grip condition.** Low-effort (5% MVC) and high-effort (40% MVC) trials are analyzed separately. At 5% MVC, target-normalized deviations are dominated by precision noise near zero, inflating apparent tremor

without indexing meaningful instability. At 40% MVC, the same spectral features carry a direct interpretable sign. Primary inference is organized around the high-effort arm; low-effort and pooled models are reported as comparisons.

**Behavioral exclusions.** Trial-level exclusions followed criteria established in Sun AY, Dastgheib M, Hu X, et al [45]. Runs were excluded if more than 90% of same/different responses were to a single button (3.62% of trials removed). Trials with missed same/different responses (1.69%), implausibly fast responses ( $< 200$  ms after the response prompt, which falls 450 ms after probe offset; 1.37%), and failures to maintain target grip force for at least two-thirds of the 2–3 s post-cue window (30–55% MVC for high grip; 1–15% MVC for low grip; 4.00%) were also removed. In total, 90.01% of trials were retained after behavioral cleaning. For the spectral analysis, additional quality screens required that each retained epoch contain no force-clipping artifacts, no missing samples, and sufficient duration for the Welch PSD estimation.

**Grip-adherence metric.** To quantify compliance with the grip manipulation, we computed a Cohen's  $d$ -analogue separability metric: the difference between the mean force observed on high-grip trials and the mean force on low-grip trials, divided by the pooled standard deviation. The first 2 s of each trial were excluded from this calculation to allow participants to reach the target. Higher values indicate greater separation between the force distributions and better adherence to the instructed arousal manipulation. This metric is used as a covariate in secondary analyses and as a manipulation check.

## Statistical analysis

**Mixed-effects models.** Perceptual accuracy (correct vs. incorrect) was modeled as a binary outcome using trial-level mixed-effects logistic regression estimated in R with the `lme4` package. The primary predictors were log band-power in the visuomotor tracking band and the physiological tremor band, entered simultaneously. Perceptual difficulty was represented by orthogonal polynomial contrasts through the fourth order, estimated from the five probe levels within each modality. This preserves the nonlinear psychometric shape of the accuracy-by-difficulty function without collapsing to coarse bins. Grip condition and task modality were included as fixed effects, along with their interactions with the force features. Random intercepts by participant accounted for stable between-subject differences in perceptual sensitivity and force regulation. Separate models were estimated

for each modality and grip condition. Primary inference focuses on the high-effort arm (40% MVC); pooled and low-effort models are reported for comparison.

**Cross-validated predictive modeling.** To assess whether continuous force features predict accuracy beyond participant-level baseline differences, subject-wise leave-one-out cross-validation was used. For each held-out participant, a model trained on the remaining participants' data generated out-of-sample predictions. Predictive accuracy was evaluated using area under the receiver operating characteristic curve (AUC). This structure prevents stable individual differences in grip strength, tremor level, or perceptual sensitivity from masquerading as trial-level cognitive-motor coupling, and provides an estimate of generalizability of the force features as biomarkers of perceptual strain.

## Results

### Behavioral load and spectral phenotype

After behavioral exclusions and spectral quality screening, the primary analysis included 64 participants contributing 6,911 high-grip trials (40% MVC; accuracy = 64.2%) and 6,671 low-grip trials (5% MVC; accuracy = 65.8%). Participants maintained high-grip targets with good fidelity (mean compliance =  $1.10\times$  instructed force; Supplementary Figure S4).

Physical load compressed perceptual sensitivity. Figure 2 shows that  $P(\text{“Different”})$  rose monotonically with stimulus level in both modalities, but high-grip trials produced lower sensitivity at intermediate levels ( $\Delta 2$ – $\Delta 3$ ), consistent with a dual-task cost concentrated where evidence is weakest. At the subject level, accuracy under high grip was lower than under low grip for 56% of participants (mean load cost = 1.3 percentage points,  $SD = 6.0$ ). The load-related accuracy drop was not accompanied by a systematic speed–accuracy trade-off (Supplementary Figure S6).

Grand-average Welch PSDs differed markedly between grip conditions (Figure 3). Under high grip, power in the 8–12 Hz tremor band was suppressed relative to low grip, whereas the 0.5–3 Hz tracking band remained elevated—reflecting active visuomotor correction around the 40% MVC target. Trial-level distributions showed that incorrect responses tended to cluster toward higher tremor power under high effort (Supplementary Figure S3).

## Inferential models and hypothesis tests

We first tested whether log band-power features improved a baseline mixed-effects logistic model of perceptual accuracy in the high-grip arm. Adding tracking- and tremor-band power did not significantly improve model fit (likelihood-ratio test:  $\chi^2(2) = 1.61$ ,  $p = 0.447$ ). In the augmented model, higher tremor power showed a numerically negative association with accuracy (OR = 0.96 (0.88, 1.04),  $p = 0.282$ ), whereas tracking power was essentially flat (OR = 1.00 (0.93, 1.08),  $p = 0.965$ ). Stimulus difficulty and modality remained strong predictors in both models (Table 1; Supplementary Table S3).

Because low- and high-grip trials reflect different motor-control regimes, we repeated the analysis separately within each grip stratum (Figure 4). Under high grip, each +1 SD increase in tremor power was associated with lower accuracy (OR = 0.96 (0.89, 1.03),  $p = 0.241$ ), consistent with the Tremor Surging Hypothesis in direction if not in significance. Under low grip, the association was similar in sign and magnitude (OR = 0.95 (0.89, 1.01),  $p = 0.114$ ), indicating that low-effort spectral noise does not show the predicted positive coupling with accuracy. The Visuomotor Collapse Hypothesis was not supported in the primary high-grip model: tracking-band power did not predict accuracy (OR = 1.00 (0.93, 1.08),  $p = 0.965$ ). Figure 5 shows the tremor  $\times$  grip interaction on marginal accuracy after adjusting for modality and difficulty.

Subject-wise cross-validation assessed whether grip-force features generalize beyond task difficulty. A random-forest classifier using stimulus difficulty and modality alone achieved out-of-sample AUC = 0.850; adding DSP features raised AUC to 0.855 ( $\Delta = 0.005$ ). The increment was small but consistent, indicating that continuous force dynamics carry incremental—not dominant—predictive information about trial-level accuracy (Supplementary Figure S7).

In an exploratory analysis linking MRI-derived locus coeruleus (LC) microstructure to load-related motor reactivity ( $N = 55$ ), LC fICVF showed a directional trend toward larger high-minus-low grip increases in tremor power among participants with lower integrity ( $r = -0.19$ , 95% bootstrap CI [-0.38, 0.07], interval including zero). This pattern is consistent with, but does not confirm, a neuromodulatory boundary condition on motor reactivity under load. The association between LC fICVF and subject-level load cost was weaker and not reliable ( $r = -0.12$ , 95% CI [-0.38, 0.18]; Supplementary Figure S8).

Together, the results offer qualified support for the Tremor Surging Hypothesis under high physical load: tremor-band power trended in the predicted negative direction, although trial-level coefficients did not reach conventional significance after controlling for difficulty and subject structure. The Visuomotor Collapse Hypothesis was not supported. Cross-validated prediction confirmed a modest incremental gain from force features beyond nominal task labels ( $\Delta\text{AUC} = 0.005$ ), consistent with a subtle rather than dominant motor signature. Exploratory LC analyses yielded a non-significant directional trend warranting higher-powered replication.

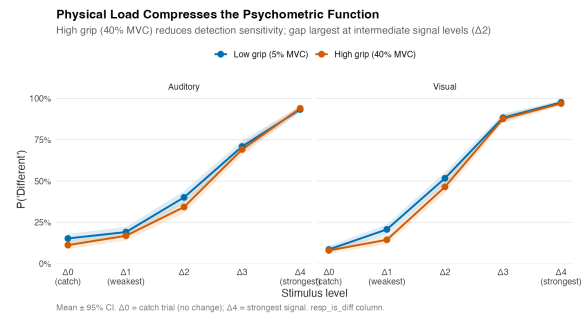


Figure 2: Physical load compressed the psychometric function, especially at intermediate stimulus levels ( $\Delta 2$ – $\Delta 3$ ). Mean  $\pm$  95% CI; catch trial =  $\Delta 0$ .

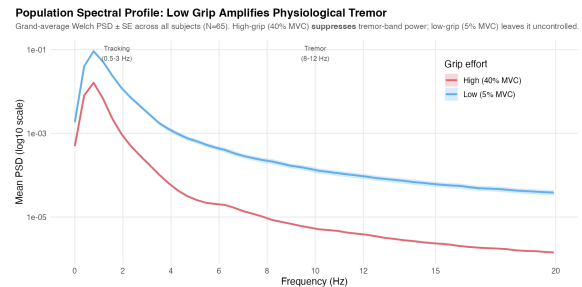


Figure 3: Grand-average Welch PSD  $\pm$  SE by grip condition. Shaded bands mark the tracking (0.5–3 Hz) and tremor (8–12 Hz) analysis windows.



Opposite Signs: Tremor Means Something Different Under Load

Left: population-level accuracy vs tremor (visual task, medium difficulty). Right: odds-ratio forest plot.

High grip OR < 1 (more tremor → more errors). Low grip OR > 1 (precision-noise artifact of near-zero 5% MVC target).

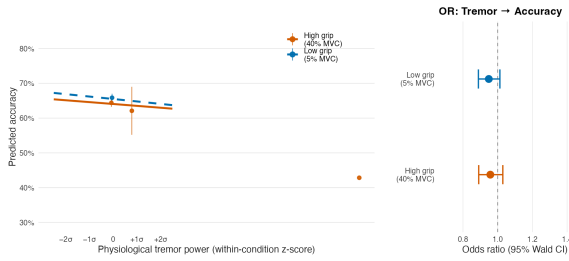


Figure 4: Stratified tremor-accuracy associations under high (40% MVC) versus low (5% MVC) grip. Left: population-level predicted accuracy across within-stratum tremor z-scores from mixed-effects models with modality and difficulty covariates (predictions averaged across auditory and visual tasks and all stimulus levels); points show binned observed accuracy. Right: odds-ratio forest plot for tremor power.

High-Frequency Motor Instability Predicts Perceptual Errors Under Physical Load

Marginal accuracy from mixed-effects model (modality + difficulty adjusted; population-level predictions)

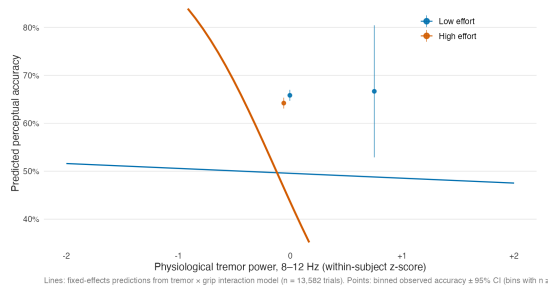


Figure 5: Marginal accuracy as a function of tremor-band power, split by grip effort. Lines show population-level fixed-effect predictions from a tremor × grip interaction model with modality and orthogonal difficulty contrasts (all stimulus levels); points show binned observed accuracy ± 95% CI (bins with  $n \geq 25$ ).

Table 1: Mixed-effects logistic regression of perceptual accuracy — high-grip (40% MVC) trials. Odds ratios (OR) with 95% Wald confidence intervals; random intercepts for participant.

| Predictor                       | OR (95% CI)          | p      |
|---------------------------------|----------------------|--------|
| Intercept                       | 1.99 (1.61, 2.46)    | < .001 |
| Modality (visual vs. auditory)  | 1.70 (1.47, 1.96)    | < .001 |
| Stimulus difficulty (linear)    | 4.76 (3.86, 5.87)    | < .001 |
| Stimulus difficulty (quadratic) | 30.19 (24.85, 36.68) | < .001 |
| Stimulus difficulty (cubic)     | 0.16 (0.14, 0.19)    | < .001 |
| Stimulus difficulty (4th order) | 1.75 (1.54, 2.00)    | < .001 |
| Tracking power 0.5–3 Hz (z)     | 1.00 (0.93, 1.08)    | 0.965  |
| Tremor power 8–12 Hz (z)        | 0.96 (0.88, 1.04)    | 0.282  |

## Discussion

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